Capstone Project: Written Responses

Question 1: Problem Definition

Describe the problem you chose to work on and why it is worth solving.
In your response:

- Clearly define the problem or question you are addressing
- Describe the context
- Identify who benefits from solving the problem and why
- Explain what a useful outcome would look like

Problem Statement

This project aims to identify patterns of criminal activity and group localities within Atlantic Canadian provinces based on similarities in crime trends between 2020 and 2024.

The central question guiding this analysis is:

Can meaningful patterns in crime data be identified that allow Atlantic provincial localities to be grouped according to similar criminal profiles and trends?

Rather than focusing on individual crime incidents, this project examines aggregated crime statistics across multiple offence categories (e.g., violent, property, drug-related offences). The goal is to understand structural similarities and differences among localities within the provinces over time.

Context

Publicly available crime data provides detailed counts of Criminal Code violations across provinces and years. While raw counts are informative, they do not immediately reveal broader patterns or relationships between regions.

From 2020 to 2024, Atlantic Canadian provinces may have experienced shifts in crime types due to social, economic, and policy-related factors. However, without systematic analysis, it is difficult to determine whether provinces share similar crime structures or whether each province follows a distinct pattern.

By applying data processing techniques and unsupervised machine learning (K-means clustering), this project transforms raw crime counts into groupings that highlight structural similarities across provinces.

Source: Statistics Canada, Table 35-10-0178-01, “Criminal Code violations, by province and territory, 2020–2024”, https://www150.statcan.gc.ca/t1/tbl1/en/cv.action?pid=3510017801

Who Benefits and Why

Several stakeholders could benefit from this analysis:

- **Policy makers and public safety agencies**: Identifying provinces with similar crime patterns can support shared policy strategies and resource allocation decisions.
- **Researchers and analysts**: Grouping provinces based on data-driven patterns enables comparative research and more targeted hypothesis development.
- **Community stakeholders**: Understanding structural crime profiles can improve transparency and inform public discussions about safety trends.

What a Useful Outcome Looks Like

A useful outcome would include:

- Clearly defined clusters of localities within Atlantic provinces based on crime composition and trends.
- Visualizations showing how localities compare across major crime categories.
- Interpretation of what distinguishes each cluster.
- A concise explanation of patterns that are not immediately visible from raw data alone.

Question 2: Approach & Tool Selection

Explain how you decided to approach the problem and which tools or techniques you selected.

In your response:

- Outline your overall approach at a high level
- Identify the specific tools, methods, or techniques used from the course
- Explain why these choices were appropriate for the problem
- Briefly note any alternative approaches you considered and why they were not selected

Overall Approach

To identify patterns of criminal activity across localities within Atlantic Canadian provinces (2020–2024), the following approach was adopted:

1. **Data Processing and Exploration:** Cleaned the dataset, removed irrelevant rows (including entries filtered using predefined patterns), handled missing values, and structured columns for analysis. Initial inspection of the dataset was performed to verify structure and data quality.

2. **Feature Engineering and Normalization:** Transformed raw counts into proportions for comparability; applied RobustScaler to standardize features for K-means clustering.

3. **Unsupervised Machine Learning (Clustering):** Applied K-means clustering to identify structural similarities, analyzing absolute crime levels, proportional composition, and year-to-year growth patterns. Optimal cluster numbers selected using Silhouette score (evaluating a range of cluster counts and selecting the configuration with the highest score).

4. **Interpretation and Reporting:** Cluster centroids and PCA plots interpreted for distinguishing characteristics of each group.

Tools, Methods, and Techniques Used

Programming Environment

Python

- Pandas: Data manipulation and cleaning
- NumPy: Numerical operations
- Matplotlib: Adjustments and custom plot details.
- Seaborn: Main visualizations for style and ease.
- Scikit-learn: Machine learning (K-means, PCA, scaling)

Data Processing Techniques

- Data cleaning and restructuring

- Aggregation
- Feature engineering (proportions and growth rates)
- Feature scaling

Machine Learning Technique

- K-means clustering
- Silhouette score for validation
- PCA for visualization

Generative AI Usage

- **Idea Generation and Research:** ChatGPT was used in the initial phase to refine the research question, explore relevant datasets, and assess the feasibility of applying unsupervised learning techniques. Suggested ideas were critically evaluated, and official data from Statistics Canada was selected to ensure reliability and methodological consistency.
- **Coding and Development:** Generative AI was used as a support tool to improve code organization, readability, and efficiency. It assisted in structuring the workflow, refining variable names, and suggesting improvements for data transformation and visualization. All AI-generated code was manually reviewed, tested, and adjusted. Continuous human oversight was required to validate outputs, ensure logical consistency, and maintain alignment with the project objectives. Analytical judgment guided all methodological decisions.
- **Documentation:** ChatGPT supported the refinement of code comments and written explanations, helping improve clarity, structure, and formal tone while preserving the technical accuracy of the analysis.

Why These Choices Were Appropriate

- Unsupervised learning is suitable since there is no target variable.
- K-means provides interpretable centroids; scaling avoids dominance by large-magnitude features.

- Proportional transformation ensures clusters reflect structure, not locality size.
- Silhouette score evaluates cluster separation and cohesion.
- Generative AI used as support tool.

Alternative Approaches Considered

- **Regression (Predictive Modeling):** Forecasting future crime totals was not feasible due to only five years of data and absence of explanatory variables.
- **Classification:** Artificial labels (e.g., “high-risk” or “low-risk”) would require subjective thresholds and could introduce bias; the objective was exploratory.
- **Hierarchical Clustering:** Offers dendrograms and flexible cluster structures but provides limited added value over K-means for centroid interpretation in this structured dataset.
- **DBSCAN (Density-Based Clustering):** Not suitable because it clusters based on density and outliers. Parameter tuning is difficult in high-dimensional structured data, and most localities are similar, making density-based detection ineffective. K-means provides more interpretable centroids.
- **Elbow Method:** Not used because within-cluster variation is minimal due to highly similar localities; the “inflection point” is ambiguous.
- **Calinski-Harabasz Index:** Not used because most localities are structurally similar, so the ratio of between-cluster to within-cluster variance does not yield meaningful differentiation.

Question 3: Reflection

Write a reflection on what you learned by completing this project.

In your response:

- Discuss what worked better or worse than expected
- Identify challenges or limitations
- Explain what you would change or improve next time
- Describe what this project taught you about selecting and applying tools to real problems

What Worked Better Than Expected

- Clustering effectively revealed structural differences among localities within Atlantic provinces.
- Meaningful groupings emerged once the dataset was cleaned, standardized, and transformed into proportional features.
- Visualizations (heatmaps, PCA scatter plots) facilitated interpretation of cluster centroids.
- Generative AI helped structure documentation and written summaries efficiently.

Challenges and Limitations

- **Cluster Concentration:** Most localities fall into a single cluster due to structural similarity, limiting differentiation. The second cluster contains only localities with slightly distinct patterns.
- **Dataset Scope:** Only aggregated crime counts are available; no demographic or socioeconomic variables are included.
- **Data Restructuring:** Transposition and aggregation were required for clustering, increasing preprocessing complexity.
- **Feature Sensitivity:** Scaling and proportional transformations were essential to avoid size-dominated clusters.

- **Cluster Validation:** Even with Silhouette scoring, small variations in feature representation can influence cluster assignments in a dataset with highly similar localities.
- **Data Access Constraints**: Downloading longer time series from Statistics Canada was limited because CSV export was unavailable for large datasets. Alternative formats intended for database use were required, increasing preprocessing complexity.

What I Would Improve Next Time

- Include external variables (population, income, unemployment) to enhance context.
- Normalize crime counts per capita for comparability.
- Experiment with alternative clustering methods (hierarchical, DBSCAN) to evaluate consistency.
- Perform stability testing on cluster assignments.

Lessons Learned About Tool Selection and Application

- Tool choice must align with data structure and analysis objectives.
- Supervised methods were inappropriate; clustering revealed patterns without predefined labels.
- Data cleaning, feature creation, and scaling were critical for meaningful results.
- Analytical interpretation was essential for connecting clusters to real-world insights. Generative AI proved most effective as a support tool, requiring human oversight for methodological accuracy and interpretation.